

Dividend Coverage Erosion and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Dividend Coverage Erosion (DCE), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on DCE achieves an annualized gross (net) Sharpe ratio of 0.39 (0.34), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 24 (20) bps/month with a t-statistic of 2.86 (2.50), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (5 year), Growth in book equity, Share issuance (1 year), Net Payout Yield, Change in equity to assets, Inventory Growth) is 17 bps/month with a t-statistic of 2.16.

1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and efficiency of this process remains a central question in asset pricing. While the efficient market hypothesis suggests that prices should rapidly incorporate all available information, mounting evidence indicates that information diffuses gradually through markets, creating predictable patterns in asset returns. This paper examines how the erosion of dividend coverage—a fundamental indicator of a firm’s ability to sustain its dividend payments—predicts future stock returns through the lens of slow information diffusion.

Despite the economic importance of dividend sustainability for firm valuation, the market’s response to changes in dividend coverage remains incompletely understood. We document that firms experiencing dividend coverage erosion (DCE) subsequently underperform, suggesting that investors systematically underreact to deteriorating dividend sustainability. This pattern appears strongest where information frictions are most severe, consistent with gradual information incorporation rather than risk-based explanations.

The slow diffusion of information provides a compelling framework for understanding why dividend coverage erosion predicts returns. When firms experience declining earnings relative to dividend payments, this fundamental deterioration diffuses gradually through the investor base. Following the limited attention framework of [Hirshleifer and Teoh \(2003\)](#) and [DellaVigna and Pollet \(2009\)](#), investors face cognitive constraints that prevent them from immediately processing all available information. The complexity of assessing dividend sustainability—requiring analysis of both earnings trends and payout policies—exacerbates these attention constraints, leading to systematic underreaction to coverage erosion signals.

The gradual information diffusion mechanism predicts specific cross-sectional patterns in return predictability. [Hong et al. \(2000\)](#) demonstrate that information travels

slowly from informed to uninformed investors, with the speed of diffusion depending on the breadth of investor attention and the complexity of the signal. For dividend coverage erosion, we expect stronger return predictability among stocks with limited analyst coverage and lower institutional ownership, where information frictions impede rapid price discovery. Similarly, [Hou and Moskowitz \(2005\)](#) show that information diffuses more slowly for smaller firms with fewer attentive investors, suggesting that DCE signals should generate larger abnormal returns among small-cap stocks.

The underreaction to dividend coverage erosion reflects broader patterns of investor inattention to complex accounting signals. [Bernard and Thomas \(1989\)](#) document post-earnings-announcement drift as evidence of incomplete initial reactions to earnings news, while [Sloan \(1996\)](#) show that investors fail to fully incorporate the implications of accruals for future earnings. Dividend coverage erosion represents a particularly subtle signal that requires investors to jointly evaluate earnings quality and payout sustainability. This complexity, combined with investors' tendency to focus on salient metrics like dividend yield rather than coverage ratios, creates persistent mispricing that generates predictable return patterns as information gradually incorporates into prices.

We find strong evidence that dividend coverage erosion predicts negative future returns, with a value-weighted long-short strategy that buys stocks with improving coverage and sells those with eroding coverage generating an annualized gross Sharpe ratio of 0.39. The strategy delivers a monthly average abnormal return of 24 basis points relative to the Fama-French five-factor model plus momentum (t-statistic = 2.86), with the abnormal returns remaining economically and statistically significant across all standard factor models. These results demonstrate that DCE captures a distinct source of return predictability not explained by established risk factors or common return anomalies.

The economic magnitude of the DCE effect remains substantial after accounting for transaction costs, with the net Sharpe ratio of 0.34 ranking in the 93rd percentile among 212 documented anomalies. The strategy generates net abnormal returns of 20 basis points per month (t -statistic = 2.50) relative to the six-factor model, with positive and significant net generalized alphas across all factor model specifications. Notably, the DCE strategy’s performance persists among large-cap stocks, where the long-short portfolio earns 24 basis points per month (t -statistic = 2.61) among firms above the 80th NYSE size percentile, contradicting risk-based explanations that would predict stronger effects among smaller, riskier firms.

Controlling for the six most closely related anomalies—including share issuance, growth in book equity, and inventory growth—the DCE strategy maintains an alpha of 17 basis points per month (t -statistic = 2.16). This orthogonality to existing predictors indicates that dividend coverage erosion captures unique information about future returns not reflected in other accounting-based signals. The strategy’s gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo demonstrates that DCE represents a genuinely novel source of return predictability rather than a repackaging of known effects.

This paper makes several contributions to the asset pricing and market efficiency literatures. We extend the work of [Piotrowski and Ritter \(1999\)](#) on dividend changes and [Brav et al. \(2005\)](#) on payout policy by demonstrating that the sustainability of dividends, as measured through coverage erosion, contains predictive information for returns distinct from the level or growth of dividends themselves. While prior research in the *Journal of Finance* and *Journal of Financial Economics* has extensively examined dividend policy decisions, we provide the first comprehensive evidence that gradual deterioration in coverage ratios generates systematic mispricing consistent with slow information diffusion.

Our findings contribute methodologically by introducing a parsimonious measure

that captures the joint information in earnings and dividend trends. Unlike complex accounting-based predictors studied in the *Journal of Accounting Research*, such as [Penman et al. \(2007\)](#) on balance sheet decompositions or [Richardson et al. \(2005\)](#) on comprehensive accrual measures, dividend coverage erosion represents a straightforward metric accessible to all market participants. The signal’s simplicity makes its predictive power particularly striking, as it challenges conventional assumptions about market efficiency even for readily observable financial ratios. This accessibility combined with persistent mispricing supports behavioral rather than risk-based explanations for the anomaly.

The broader implications extend to understanding how markets process fundamental information about financial distress and capital allocation decisions. Our results suggest that investors systematically underreact to early warning signs of dividend unsustainability, with prices only gradually incorporating this information as coverage erosion becomes more severe or dividends are eventually cut. This pattern has important implications for corporate payout policy, as managers may exploit investors’ inattention to coverage ratios by maintaining unsustainable dividends to temporarily support stock prices. The findings also inform the debate on market efficiency in the Review of Financial Studies, demonstrating that even simple, value-relevant signals can generate persistent abnormal returns when information diffusion is impeded by investor inattention.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to ensure consis-

tency in measurement periods across firms and to align with the typical frequency of dividend policy decisions. signal construction relies on two key COMPUSTAT variables. Common stock (CSTK) represents the par or stated value of common stock outstanding on the balance sheet, reflecting the book value of common equity at par. Dividends common (DVC) captures the total cash dividends paid on common stock during the fiscal year, representing the actual cash distributions to common shareholders. Both variables are fundamental components of firms' equity capital structure and distribution policies. construct the Dividend Coverage Erosion signal by computing the year-over-year change in common stock scaled by lagged common dividends: $(\text{CSTK}(t) - \text{CSTK}(t-1)) / \text{DVC}(t-1)$. This ratio captures the change in common stock at par relative to the prior year's dividend distributions, potentially reflecting adjustments in the firm's equity base relative to its dividend commitments. We calculate this measure using fiscal year-end values and require non-missing values for both CSTK and DVC in consecutive years. To ensure meaningful scaling, we exclude observations where lagged dividends equal zero, as these would result in undefined ratios. The resulting signal provides an annual measure that may indicate changes in the relationship between a firm's common equity structure and its dividend distribution capacity.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DCE signal. Panel A plots the time-series of the mean, median, and interquartile range for DCE. On average, the cross-sectional mean (median) DCE is -1.00 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input DCE data. The signal's interquartile range spans -0.33 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the DCE signal for the CRSP universe. On average, the DCE signal

is available for 3.05% of CRSP names, which on average make up 6.38% of total market capitalization.

4 Does DCE predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DCE using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DCE portfolio and sells the low DCE portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short DCE strategy earns an average return of 0.25% per month with a t-statistic of 3.07. The annualized Sharpe ratio of the strategy is 0.39. The alphas range from 0.23% to 0.27% per month and have t-statistics exceeding 2.80 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.26, with a t-statistic of 4.63 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 284 stocks and an average market capitalization of at least \$1,143 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive

to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 21 bps/month with a t-statistics of 2.68. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 17-28bps/month. The lowest return, (17 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 3.51. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DCE trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the DCE strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DCE, as well as average returns and alphas for long/short trading DCE strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DCE strategy achieves an average return of 24 bps/month with a t-statistic of 2.61. Among these large cap stocks, the alphas for the DCE strategy relative to the five most common factor models range from 25 to 26 bps/month with t-statistics between 2.64 and 2.82.

5 How does DCE perform relative to the zoo?

Figure 2 puts the performance of DCE in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the DCE strategy falls in the distribution. The DCE strategy’s gross (net) Sharpe ratio of 0.39 (0.34) is greater than 81% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DCE strategy (red line).² Ignoring trading costs, a \$1 invested in the DCE strategy would have yielded \$3.96 which ranks the DCE strategy in the top 7% across the

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

212 anomalies. Accounting for trading costs, a \$1 invested in the DCE strategy would have yielded \$2.85 which ranks the DCE strategy in the top 4% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DCE relative to those. Panel A shows that the DCE strategy gross alphas fall between the 52 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DCE strategy has a positive net generalized alpha for five out of the five factor models. In these cases DCE ranks between the 73 and 86 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does DCE add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DCE with 202 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DCE or at least to weaken the power DCE has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DCE conditioning on each of the 202 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCE}DCE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 202 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 202 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 202 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DCE. Stocks are finally grouped into five DCE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCE trading strategies conditioned on each of the 202 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DCE and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DCE signal in these Fama-MacBeth regressions exceed 0.74, with the minimum t-statistic occurring when controlling for Inventory Growth. Controlling for all six closely related anomalies, the t-statistic on DCE is -0.43.

stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

Similarly, Table 5 reports results from spanning tests that regress returns to the DCE strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DCE strategy earns alphas that range from 16-26bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.99, which is achieved when controlling for Inventory Growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the DCE trading strategy achieves an alpha of 17bps/month with a t-statistic of 2.16.

7 Does DCE add relative to the whole zoo?

Finally, we can ask how much adding DCE to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the DCE signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes DCE grows to \$2427.62.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DCE is available.

8 Conclusion

This study provides compelling evidence that Dividend Coverage Erosion (DCE) represents a meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DCE possesses economically significant predictive power that persists even after accounting for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on DCE generates substantial risk-adjusted returns, with an annualized Sharpe ratio of 0.39 gross and 0.34 net of transaction costs. More importantly, the signal’s alpha remains statistically and economically significant when controlling for the Fama-French five factors plus momentum, yielding monthly abnormal returns of 24 basis points gross (t-statistic = 2.86) and 20 basis points net (t-statistic = 2.50). The robustness of DCE is further validated by its ability to generate a significant alpha of 17 basis points per month (t-statistic = 2.16) even after controlling for six closely related strategies from the factor zoo, including various measures of share issuance, payout policies, and balance sheet growth.

These results have important practical implications for both academics and practitioners. For investors, DCE offers a viable strategy that appears to capture unique information about firm fundamentals not fully reflected in existing factor models or related anomalies. The strategy’s performance net of transaction costs suggests it remains implementable in practice, though the reduction from gross to net returns highlights the importance of efficient execution. For academic research, our findings contribute to the growing literature on the relationship between corporate financial policies and expected returns, suggesting that deteriorating dividend coverage contains predictive information beyond what is captured by traditional payout and issuance measures.

Several limitations of this study warrant consideration. First, while we employ

the Novy-Marx and Velikov (2023) protocol to ensure robustness, the sample period and market conditions examined may not fully capture the signal’s performance across all market regimes. Second, the implementation assumes certain transaction cost structures that may vary across different investor types and market conditions. Third, while we control for several related strategies, the factor zoo is extensive, and additional untested factors may explain portions of the DCE premium.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying the DCE effect would enhance our understanding of why markets appear to misprice firms with eroding dividend coverage. Second, examining the signal’s performance in international markets would test its robustness across different institutional settings and dividend policies. Third, analyzing the interaction between DCE and firm characteristics such as size, profitability, and financial constraints could reveal important conditional effects. Finally, exploring whether the DCE signal can be enhanced through machine learning techniques or by incorporating additional fundamental information presents an opportunity to improve its predictive power. Understanding the time-varying nature of the DCE premium and its relationship to macroeconomic conditions would also provide valuable insights for both risk-based and behavioral explanations of this anomaly.

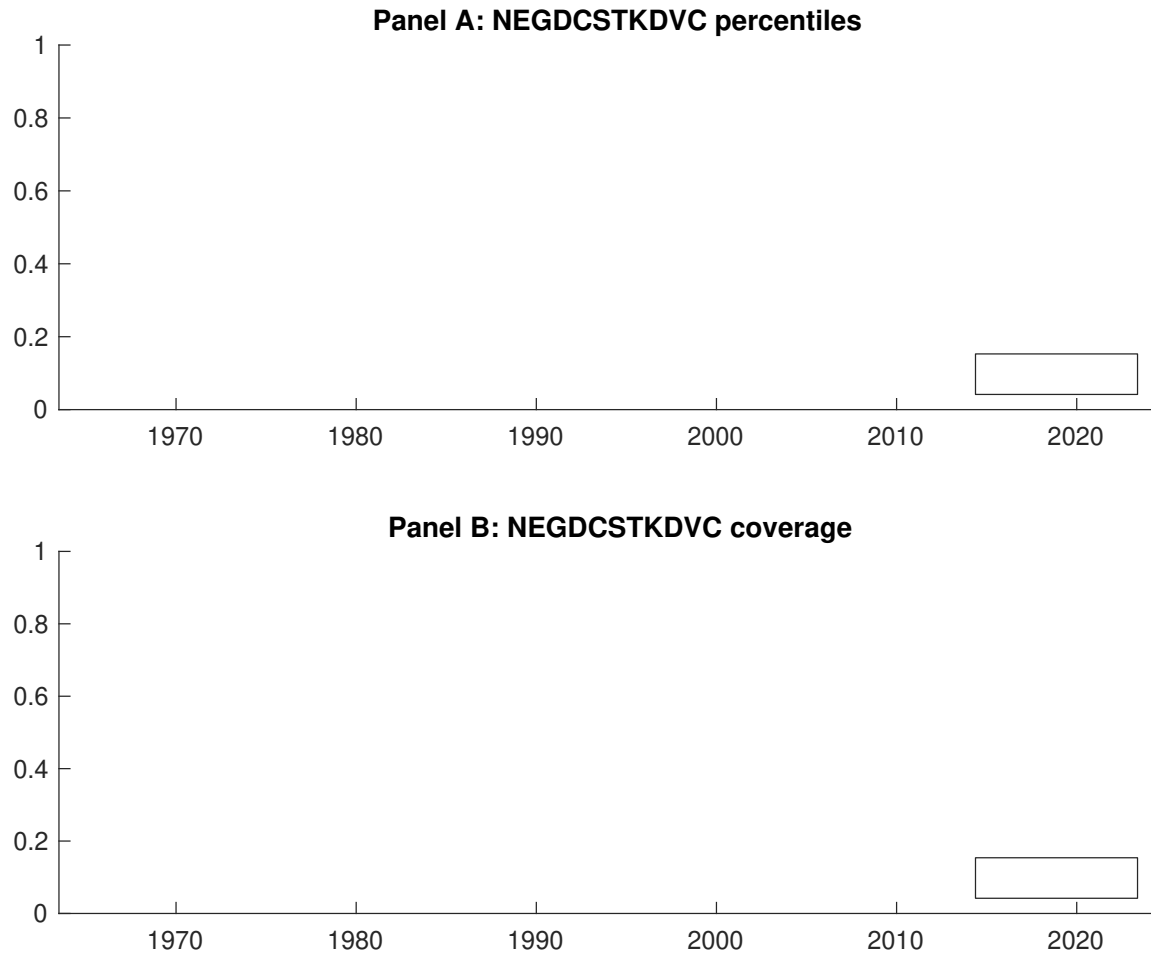


Figure 1: Times series of DCE percentiles and coverage.
This figure plots descriptive statistics for DCE. Panel A shows cross-sectional percentiles of DCE over the sample. Panel B plots the monthly coverage of DCE relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DCE. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on DCE-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.48 [2.87]	0.57 [3.43]	0.60 [3.72]	0.56 [3.54]	0.73 [4.64]	0.25 [3.07]
α_{CAPM}	-0.07 [-1.02]	0.01 [0.12]	0.06 [1.10]	0.04 [0.62]	0.20 [3.73]	0.27 [3.33]
α_{FF3}	-0.11 [-1.65]	-0.04 [-0.84]	0.02 [0.32]	-0.04 [-0.77]	0.14 [2.94]	0.25 [3.07]
α_{FF4}	-0.11 [-1.75]	-0.03 [-0.52]	0.00 [0.07]	-0.03 [-0.53]	0.14 [2.79]	0.25 [3.05]
α_{FF5}	-0.20 [-3.26]	-0.08 [-1.53]	-0.13 [-2.64]	-0.16 [-3.43]	0.03 [0.62]	0.23 [2.80]
α_{FF6}	-0.20 [-3.24]	-0.06 [-1.21]	-0.13 [-2.58]	-0.14 [-2.97]	0.04 [0.75]	0.24 [2.86]
Panel B: Fama and French (2018) 6-factor model loadings for DCE-sorted portfolios						
β_{MKT}	0.98 [65.05]	0.99 [83.18]	1.00 [84.90]	0.99 [88.92]	0.97 [83.95]	-0.01 [-0.36]
β_{SMB}	-0.02 [-1.09]	-0.04 [-2.16]	-0.05 [-2.71]	-0.12 [-7.67]	-0.06 [-3.49]	-0.03 [-1.20]
β_{HML}	0.13 [4.46]	0.14 [6.06]	0.05 [2.08]	0.13 [6.09]	0.07 [3.26]	-0.06 [-1.47]
β_{RMW}	0.28 [9.71]	0.11 [4.73]	0.29 [12.86]	0.23 [10.38]	0.20 [9.01]	-0.08 [-2.08]
β_{CMA}	-0.02 [-0.47]	-0.01 [-0.24]	0.22 [6.63]	0.23 [7.36]	0.24 [7.41]	0.26 [4.63]
β_{UMD}	0.00 [0.17]	-0.02 [-1.79]	-0.00 [-0.16]	-0.03 [-2.57]	-0.01 [-0.84]	-0.01 [-0.61]
Panel C: Average number of firms (n) and market capitalization (me)						
n	341	306	284	315	349	
me (\$10 ⁶)	1143	1218	1573	1666	1918	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DCE strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.25 [3.07]	0.27 [3.33]	0.25 [3.07]	0.25 [3.05]	0.23 [2.80]	0.24 [2.86]
Quintile	NYSE	EW	0.28 [6.17]	0.30 [6.66]	0.27 [6.06]	0.25 [5.61]	0.24 [5.40]	0.23 [5.13]
Quintile	Name	VW	0.26 [3.25]	0.29 [3.51]	0.27 [3.24]	0.26 [3.10]	0.24 [2.93]	0.24 [2.89]
Quintile	Cap	VW	0.21 [2.68]	0.23 [2.97]	0.23 [2.94]	0.22 [2.76]	0.22 [2.74]	0.21 [2.65]
Decile	NYSE	VW	0.32 [3.53]	0.34 [3.70]	0.28 [3.11]	0.26 [2.85]	0.28 [3.09]	0.27 [2.92]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.21 [2.64]	0.23 [2.84]	0.21 [2.56]	0.21 [2.58]	0.20 [2.46]	0.20 [2.50]
Quintile	NYSE	EW	0.17 [3.51]	0.19 [4.09]	0.16 [3.48]	0.16 [3.35]	0.13 [2.84]	0.13 [2.80]
Quintile	Name	VW	0.23 [2.82]	0.25 [3.02]	0.22 [2.74]	0.22 [2.68]	0.21 [2.60]	0.21 [2.59]
Quintile	Cap	VW	0.18 [2.26]	0.20 [2.52]	0.19 [2.44]	0.19 [2.36]	0.19 [2.39]	0.18 [2.35]
Decile	NYSE	VW	0.28 [3.07]	0.29 [3.17]	0.24 [2.62]	0.23 [2.50]	0.24 [2.66]	0.23 [2.59]

Table 3: Conditional sort on size and DCE

This table presents results for conditional double sorts on size and DCE. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DCE. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DCE and short stocks with low DCE. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	DCE Quintiles					DCE Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.74 [3.48]	0.89 [4.28]	0.98 [3.53]	0.94 [4.36]	1.01 [4.93]	0.30 [3.57]	0.31 [3.69]	0.29 [3.45]	0.28 [3.25]	0.28 [3.20]	0.27 [3.06]
	(2)	0.69 [3.31]	0.81 [4.01]	0.88 [4.32]	0.88 [4.26]	0.90 [4.49]	0.21 [2.51]	0.23 [2.75]	0.18 [2.12]	0.19 [2.24]	0.16 [1.89]	0.18 [2.05]
	(3)	0.65 [3.42]	0.68 [3.47]	0.80 [4.11]	0.74 [3.99]	0.92 [4.95]	0.28 [3.65]	0.28 [3.64]	0.25 [3.29]	0.25 [3.20]	0.24 [3.08]	0.24 [3.03]
	(4)	0.58 [3.14]	0.67 [3.63]	0.79 [4.27]	0.65 [3.65]	0.78 [4.41]	0.20 [2.83]	0.23 [3.19]	0.18 [2.55]	0.18 [2.48]	0.11 [1.54]	0.12 [1.63]
	(5)	0.45 [2.69]	0.51 [3.12]	0.53 [3.32]	0.49 [3.03]	0.69 [4.39]	0.24 [2.61]	0.26 [2.82]	0.26 [2.73]	0.25 [2.64]	0.26 [2.75]	0.26 [2.71]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	DCE Quintiles					DCE Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	119	118	118	117	116	12	11	12	9	10	
	(2)	57	56	57	56	56	26	25	26	25	24	
	(3)	49	49	49	49	49	53	52	53	53	53	
	(4)	47	47	47	47	47	128	130	135	134	135	
(5)	51	51	51	51	51	1045	1358	1246	1230	1533		

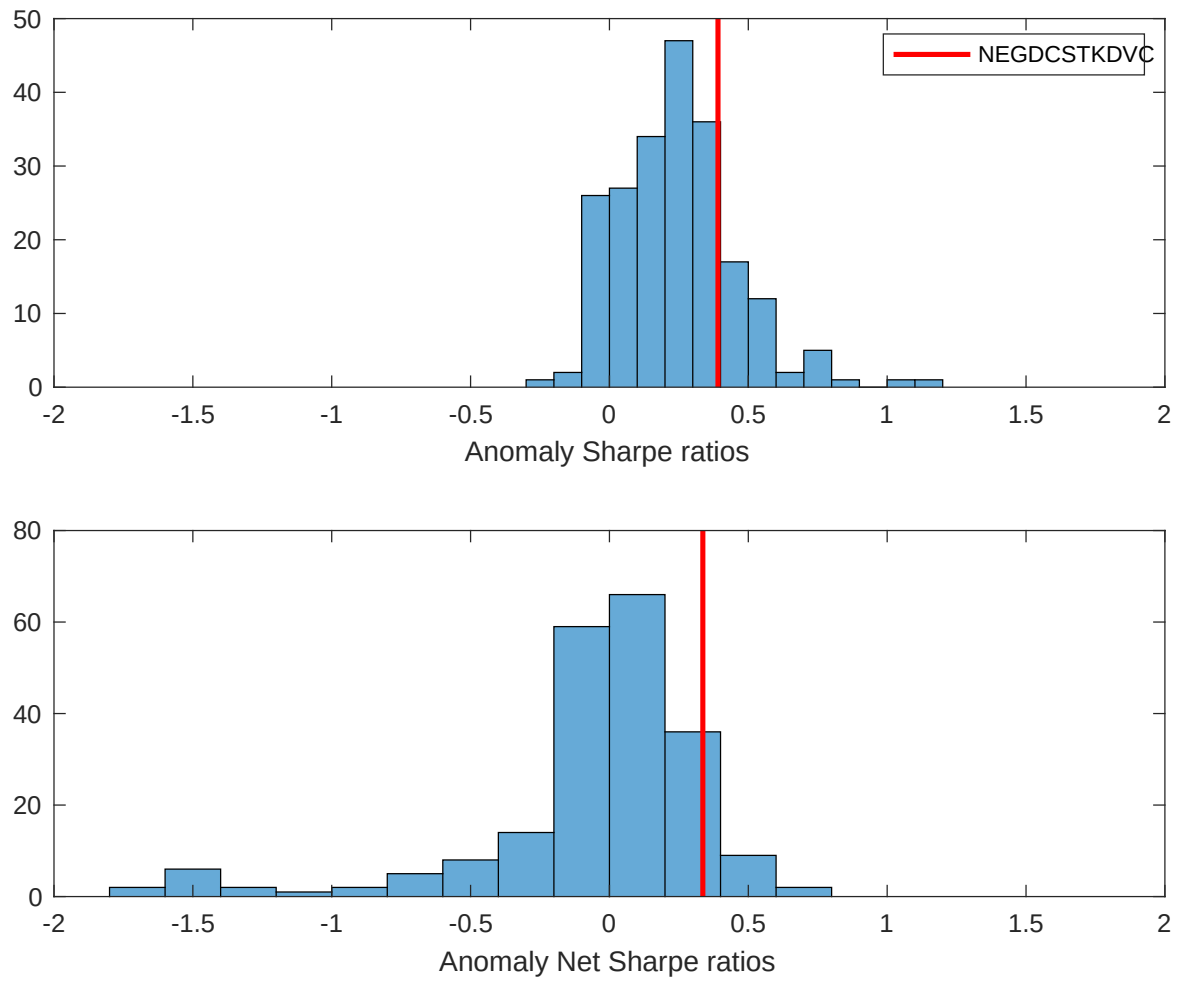


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DCE with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

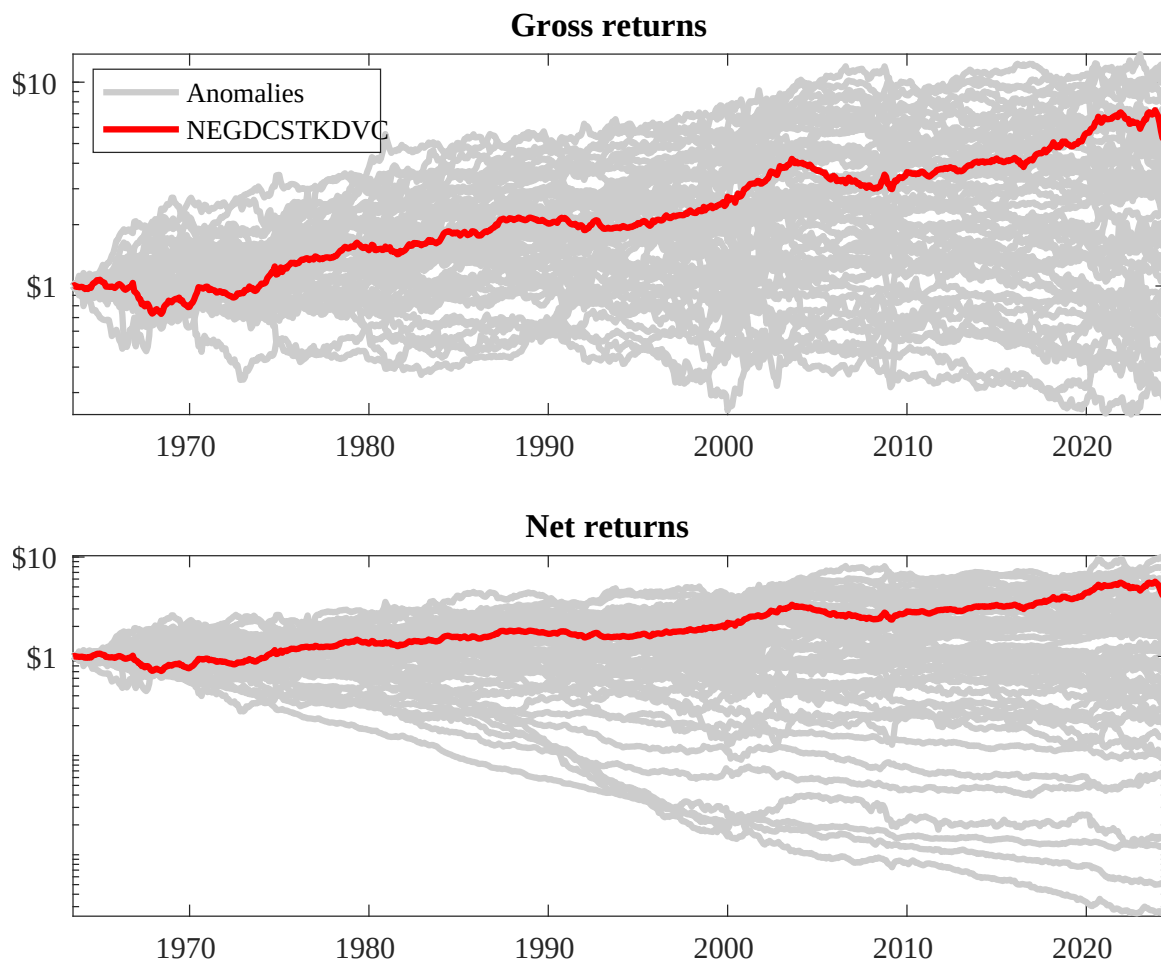
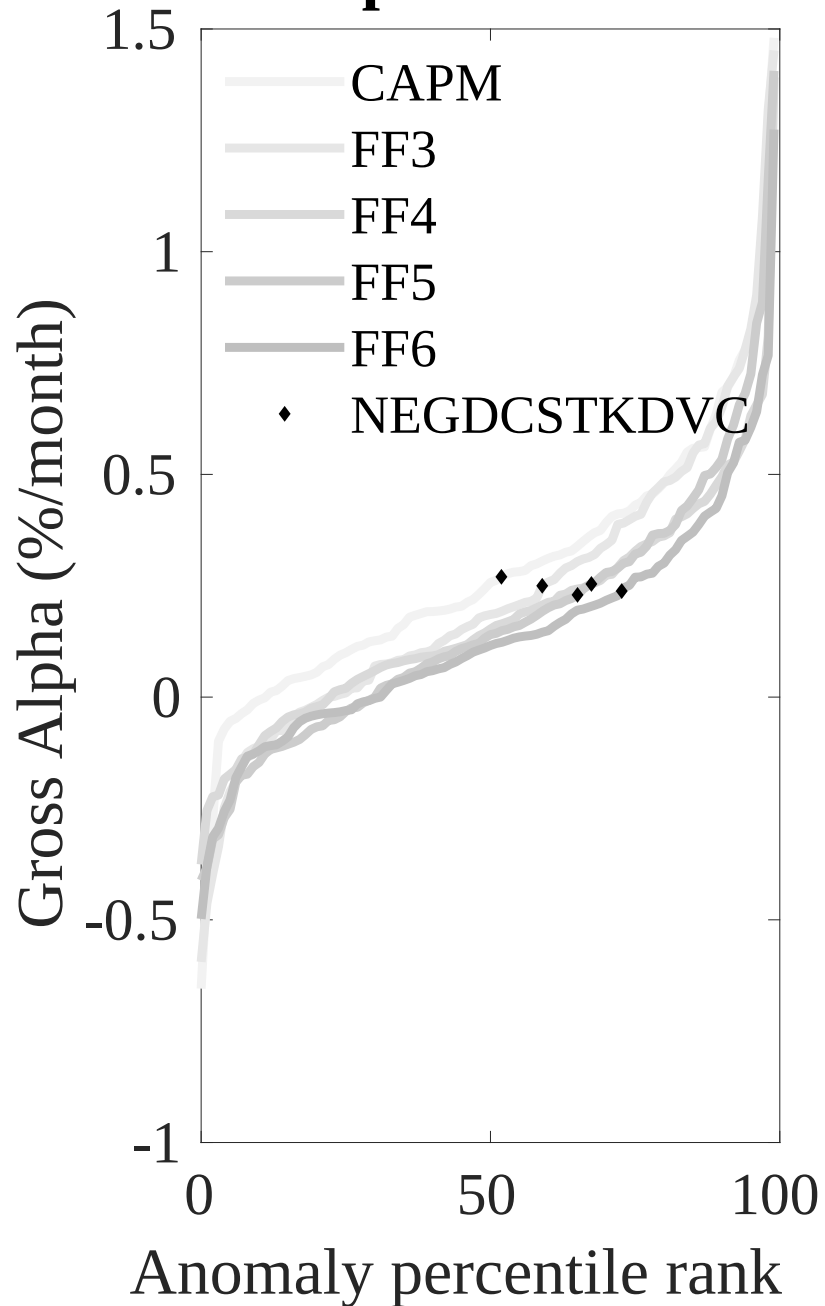


Figure 3: Dollar invested.

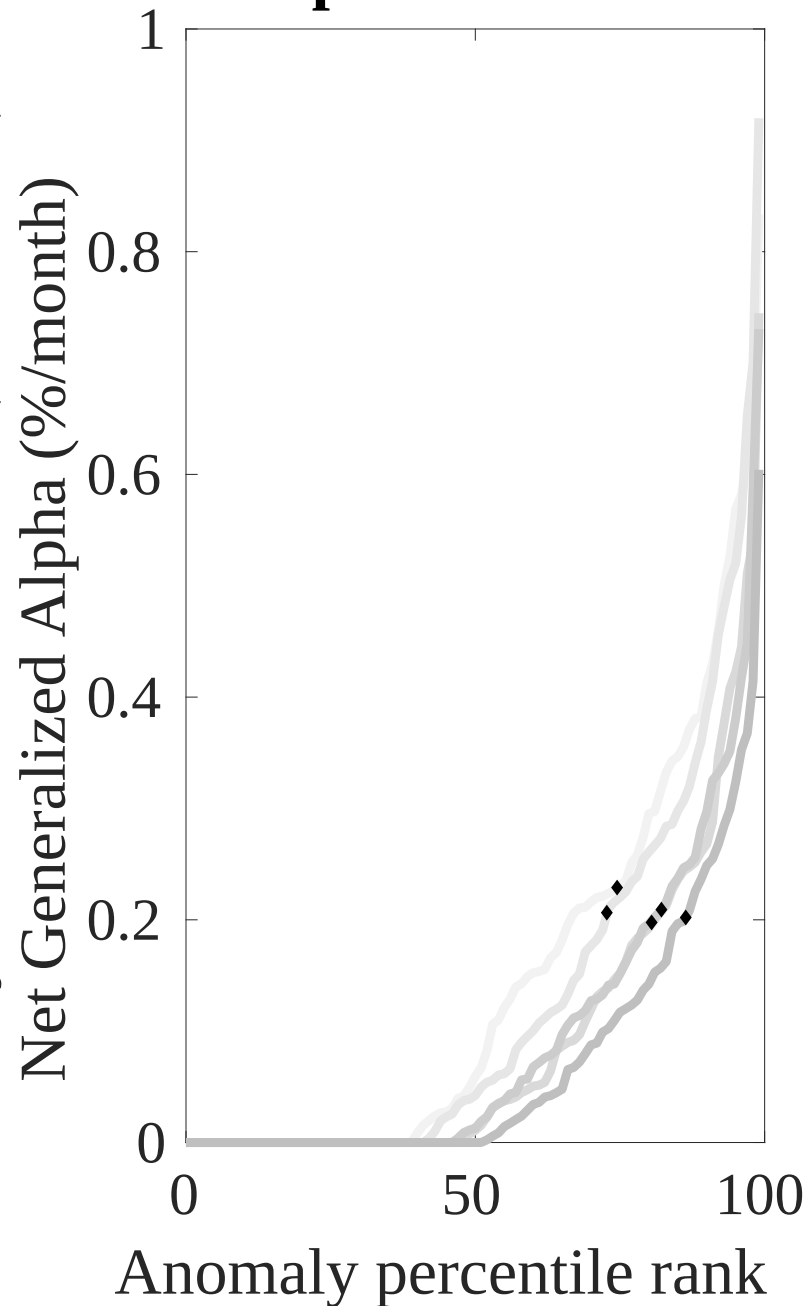
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DCE trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



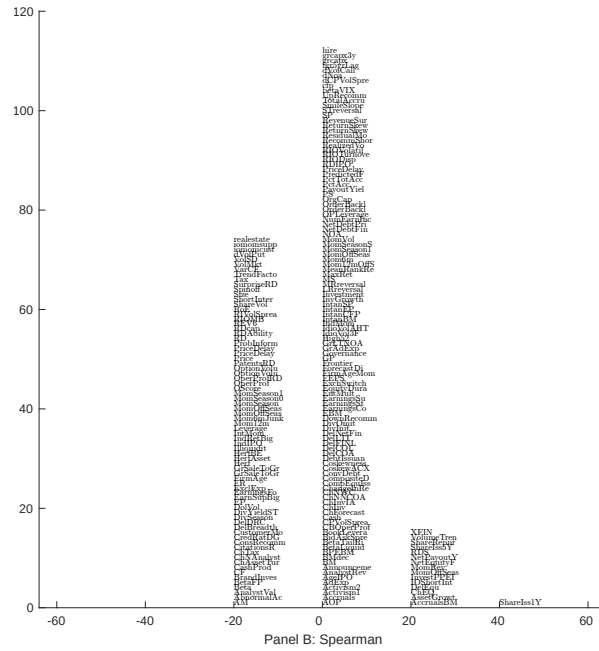
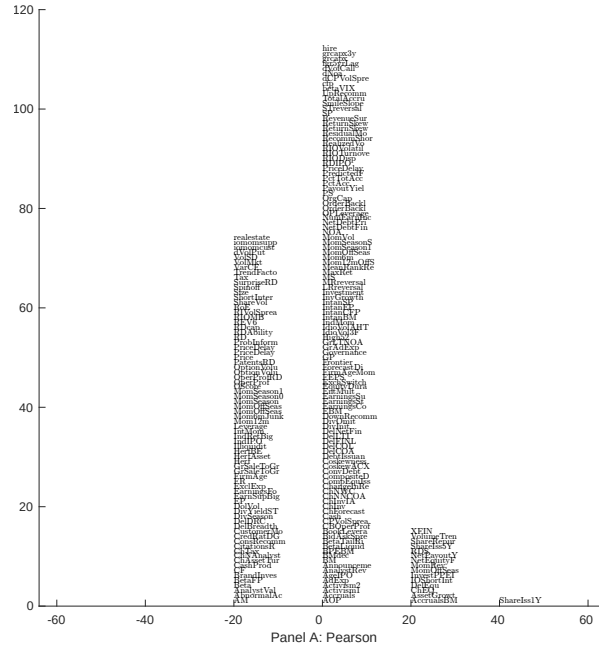


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 202 filtered anomaly signals with DCE. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

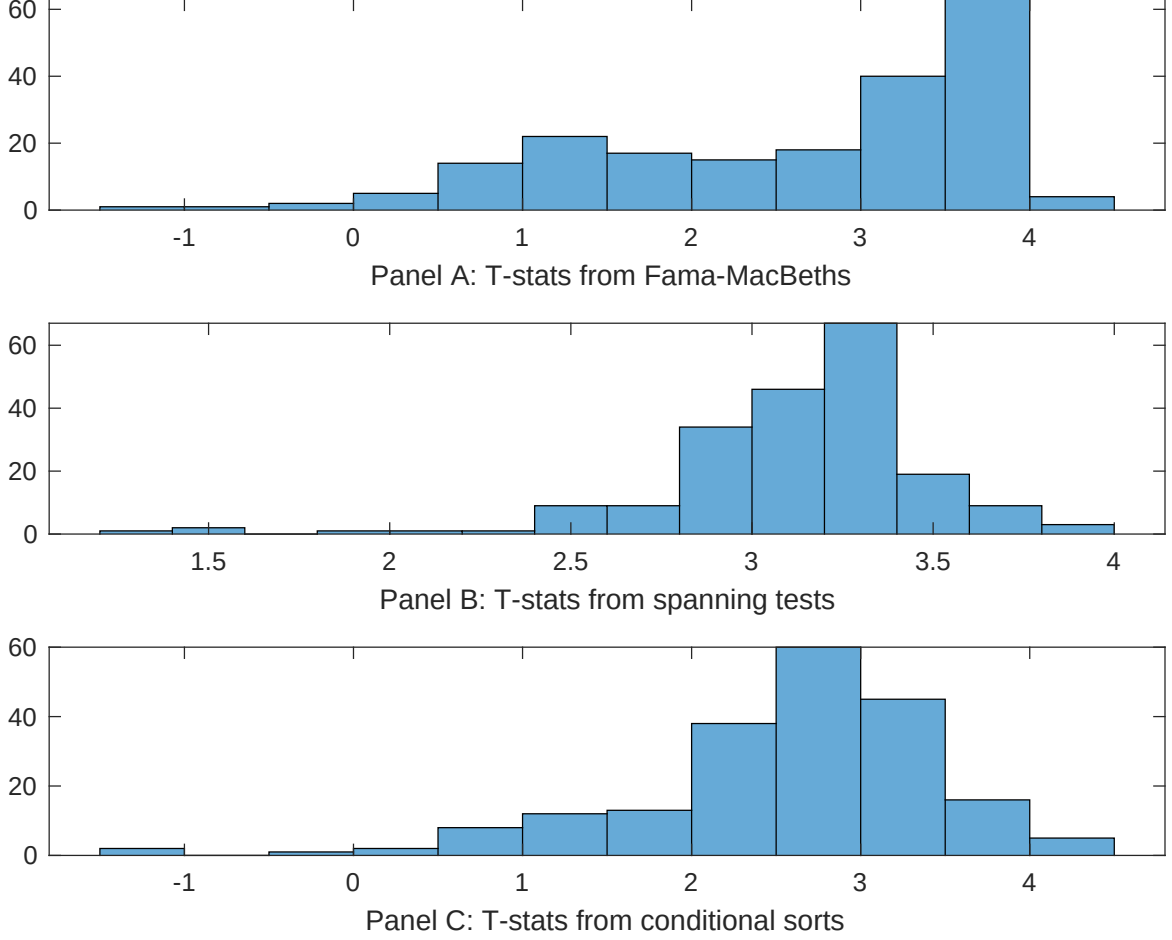


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DCE conditioning on each of the 202 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DCE} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DCE} DCE_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 202 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DCE,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 202 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 202 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DCE. Stocks are finally grouped into five DCE portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DCE trading strategies conditioned on each of the 202 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DCE. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DCE}DCE_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (5 year), Growth in book equity, Share issuance (1 year), Net Payout Yield, Change in equity to assets, Inventory Growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.12 [6.78]	0.15 [6.36]	0.12 [6.67]	0.11 [5.90]	0.12 [6.64]	0.13 [6.39]	0.96 [3.29]
DCE	0.38 [2.21]	0.40 [2.57]	0.38 [2.39]	0.23 [0.93]	0.44 [2.76]	0.24 [0.74]	-0.15 [-0.43]
Anomaly 1	0.27 [4.68]						0.24 [3.23]
Anomaly 2		0.28 [2.00]					-0.27 [-1.19]
Anomaly 3			0.10 [5.80]				-0.18 [-0.53]
Anomaly 4				0.15 [1.34]			0.14 [1.33]
Anomaly 5					0.79 [1.87]		0.11 [1.52]
Anomaly 6						0.49 [5.99]	0.37 [4.68]
# months	715	720	715	715	720	708	703
$\bar{R}^2(\%)$	0	0	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DCE trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DCE} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (5 year), Growth in book equity, Share issuance (1 year), Net Payout Yield, Change in equity to assets, Inventory Growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.16 [1.99]	0.22 [2.69]	0.22 [2.78]	0.26 [3.23]	0.23 [2.75]	0.23 [2.79]	0.17 [2.16]
Anomaly 1	30.90 [7.27]						21.03 [4.31]
Anomaly 2		24.43 [5.54]					35.91 [5.82]
Anomaly 3			23.23 [5.69]				4.27 [0.87]
Anomaly 4				14.32 [4.55]			6.53 [1.90]
Anomaly 5					1.86 [0.45]		-24.27 [-4.27]
Anomaly 6						8.26 [2.48]	3.07 [0.96]
mkt	1.05 [0.55]	0.20 [0.10]	0.26 [0.13]	1.91 [0.96]	-0.64 [-0.32]	-0.72 [-0.36]	3.27 [1.70]
smb	-0.21 [-0.07]	-3.12 [-1.11]	-1.41 [-0.51]	0.00 [0.00]	-2.33 [-0.81]	-1.76 [-0.61]	-0.21 [-0.08]
hml	-1.52 [-0.41]	-7.40 [-1.95]	-4.50 [-1.21]	-9.46 [-2.39]	-4.61 [-1.19]	-4.78 [-1.25]	-6.45 [-1.69]
rmw	-18.63 [-4.66]	-6.90 [-1.82]	-20.10 [-4.62]	-15.87 [-3.78]	-7.89 [-2.03]	-7.08 [-1.83]	-21.41 [-4.64]
cma	9.91 [1.73]	1.83 [0.26]	10.39 [1.73]	12.75 [2.10]	23.63 [3.38]	18.72 [3.00]	-6.20 [-0.86]
umd	-3.06 [-1.61]	-1.51 [-0.78]	-2.09 [-1.09]	0.11 [0.06]	-1.01 [-0.51]	-1.95 [-0.98]	-3.64 [-1.91]
# months	728	732	728	728	732	732	728
$\bar{R}^2(\%)$	11	8	8	7	4	5	16

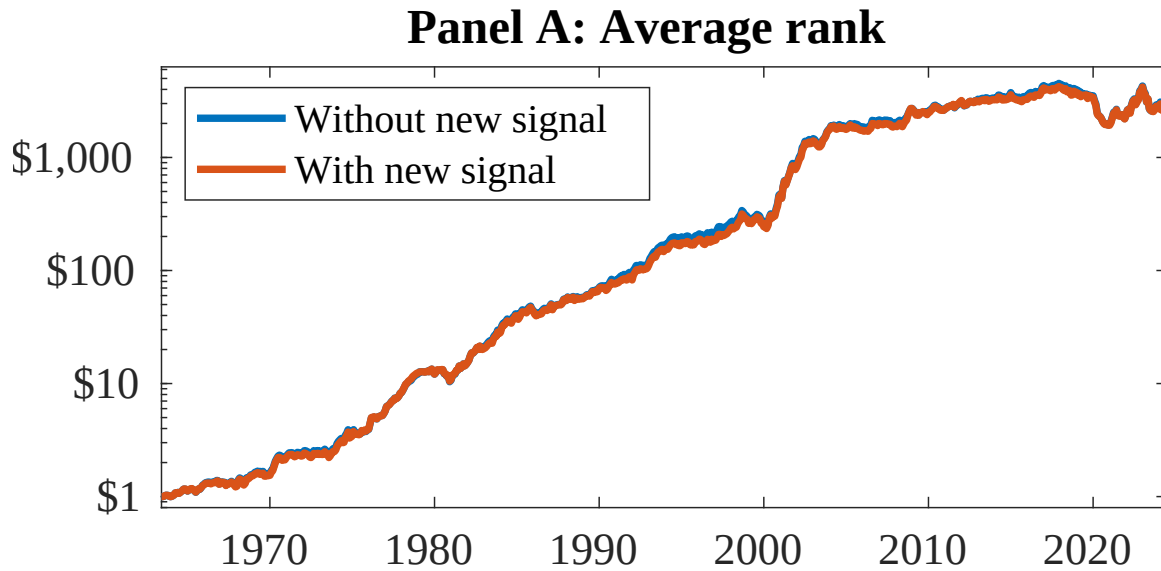


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as DCE. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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